

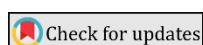


Shaping smarter prediction: A literature review of GIS, remote sensing and machine learning applications in digital soil organic carbon mapping

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Article History:

Received: 21 October 2025

Revised: 3 March 2026

Accepted: 12 March 2026

Published: 3 July 2026



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Abstract: Soil organic carbon (SOC) is an important indicator of soil health. Its regular monitoring helps achieve sustainable soil management, therefore, ensuring high agricultural productivity. The predicted SOC maps are becoming crucial and more important to monitor the climate change mitigation, sustainable environment. These particular needs have incited scientists and researchers to think about smart mapping approaches using new information technologies. This article aims to provide an overview of how GIS and remote sensing, modeled using machine learning algorithms, can contribute to SOC prediction for better understanding its spatial distribution across landscape. The review highlights the important role of Remote Sensing, Geographic Information System-Machine Learning (RS-GIS-ML) technologies. These approaches benefit from both field-collected data and features derived from spectral imagery acquired through various sensors (Multispectral, Hyperspectral, LiDAR and Radar). ML algorithms further enhance this process by modeling the complex relationships among multiple input variables, enabling the accurate SOC content in the form of spatial maps. The structure of this review is organized follow three principal sections: (1) use of remote sensing in SOC prediction, (2) machine learning methods for SOC prediction and (3) the integrated RS-GIS-ML approach for SOC mapping. The article concludes with a discussion on the future prospects of digital SOC mapping using geospatial technologies and ML algorithms.

Keywords: Soil organic carbon; remote sensing; GIS; machine learning; digital soil mapping

Citation: Khaddi, E., Zaher, H., & Naim, H. (2026). Shaping smarter prediction: A literature review of GIS, remote sensing and machine learning applications in digital soil organic carbon mapping. *Turk. J. Remote Sens.*, 8, 1808033. <https://doi.org/10.51489/tuzal.1808033>

1. Introduction

Soil organic carbon (SOC) mapping plays a critical role in understanding terrestrial carbon cycles and addressing climate change-related challenges. SOC is widely recognized as the primary indicator of soil organic matter (SOM) and is fundamental to agricultural productivity and long-term soil fertility (Zaher et al., 2020; Pouladi et al., 2023; Ermolieva et al., 2024). Accurate assessment of SOC is essential for characterizing its spatial variability and identifying appropriate land management practices. Furthermore, SOC mapping supports carbon offset initiatives and, although the impacts may be modest, can contribute to mitigation strategies such as carbon sequestration (Tajik et al., 2020; Heil et al., 2022). Conventional SOC prediction relies on field-based soil sampling, which is often labor-intensive, time-consuming, and costly (Tajik et al., 2020). In response to increasing demands within the agricultural sector, the use of remote sensing (RS) and geographic information systems (GIS) for SOC estimation has expanded, driven by their potential for improved accuracy, rapid data acquisition, and cost-effectiveness (Adhikari et al., 2020). However, SOC estimation remains a major challenge in precision agriculture, as research continues to seek methods that are rapid, accurate, and capable of generating spatially detailed, high-resolution outputs (Ye et al., 2021).

Remote sensing provides an advanced technological means of acquiring spatially continuous data by measuring surface spectral reflectance across the electromagnetic

spectrum. These observations can be used to infer various soil properties, including organic carbon content. The spatial distribution of SOC is strongly influenced by soil characteristics, topographic factors, and spectral indices, all of which can be effectively captured and analyzed using RS and GIS data (Tajik et al., 2020). Recent developments in spectroscopic sensor technology, together with high-performance computing and advanced machine learning (ML) algorithms, have substantially improved access to and processing of georeferenced data, enabling the generation of highly accurate, high-resolution SOC maps (Heil et al., 2022). Digital soil mapping (DSM) approaches that integrate RS data with ML algorithms are now widely recognized as efficient tools for SOC prediction. However, despite their advantages, RS–ML-based methods are subject to limitations and uncertainties related to data quality, model selection, and environmental variability, all of which can influence prediction accuracy. Therefore, continuous monitoring and iterative model refinement are required to adequately represent SOC spatial variability and ensure reliable predictions.

The integration of RS and GIS data with ML algorithms enables systematic analysis of complex interactions among spectral responses, environmental variables, and soil properties for spatially explicit SOC prediction (Dong et al., 2025). Consequently, RS-based predictive models and geographic representations of SOC provide valuable insights into soil spatial variability and carbon dynamics, supporting sustainable agriculture and environmental assessment (Ye et al., 2021; Mishra et al., 2022). The incorporation of spectral data together with ancillary soil and environmental attributes has been shown to significantly enhance model performance and prediction accuracy for SOC (Mishra et al., 2022; Zhu et al., 2023). Most RS data used for calibrating DSM models are derived from multispectral and hyperspectral satellite platforms, which offer extensive spatial coverage and detailed spectral information essential for reliable SOC estimation. Recent advances in RS and ML technologies present important opportunities to improve the precision and efficiency of SOC assessment, thereby enabling better agricultural management and more informed environmental policy and land-use decisions (Tajik et al., 2020; Heil et al., 2022).

Despite significant progress in integrating remote sensing, geographic information systems (GIS), and machine learning (ML) for SOC mapping, several fundamental challenges limit the robustness of these methodologies and their broader application. One of the most important challenges is data quality and representation. SOC prediction models are highly sensitive to sampling intensity, spatial heterogeneity, and spectral resolution. Variations in soil moisture, vegetation cover, surface roughness, and weather conditions can significantly alter spectral reflectance, introducing uncertainty into SOC estimates. Furthermore, spatial and temporal variability among multi-source datasets such as satellite imagery, topographic derivatives, and auxiliary environmental variables poses challenges for data standardization and model integration. The generalizability and transferability of ML models across different agroecological regions also remain limited, as models calibrated under specific environmental conditions often exhibit lower predictive performance when applied to new areas. Additionally, reliance on complex and opaque algorithms may hinder transparency, reduce stakeholder confidence and limit practical application. Finally, the lack of standardized protocols for SOC modeling, validation, and uncertainty assessment complicates inter-study comparisons and impedes the development of consistent monitoring frameworks essential for climate change mitigation and sustainable land management.

The structure of this paper is organized as follows. Section 2 provides an overview of remote sensing approaches for SOC prediction, including RS technologies, soil spectral characteristics, and environmental covariates. Section 3 reviews machine learning methods commonly applied to SOC prediction. Section 4 evaluates integrated RS–GIS–ML approaches for SOC estimation and their predictive performance. Section 5 presents a critical discussion of the advantages, limitations, and future research directions. Finally, Section 6 concludes the study.

2. Use of Remote Sensing for Soil Organic Carbon Prediction

Remote sensing (RS) is a powerful technology used to capture information about the Earth's surface from a distance. It involves acquiring data without direct physical contact with a specific area. This is made possible by the various sensors mounted on the RS platforms. RS technology has revolutionized Digital Soil Mapping (DSM) by providing spatially continuous data layers that can be used to predict and map soil properties. These technologies have become invaluable for the efficient and spatially extensive estimation and monitoring of Soil Organic Carbon (SOC) stocks, fulfilling key requirements in environmental management and sustainable agriculture (Castaldi et al., 2019). Accurate prediction of organic carbon content at the global scale is essential for monitoring carbon stocks, evaluating the impacts of climate change, and developing national agricultural policies that consider organic carbon as a major factor in mitigating greenhouse gases and sustaining agricultural productivity (Xiao et al., 2024).

2.1. Remote sensing technologies

To date, remote sensing technologies have revolutionized Earth sciences by providing fundamental insights into Earth systems and processes, including soil science, terrestrial carbon prediction (Li et al., 2024). So those mentioned remote sensing techniques for terrestrial carbon prediction are among the most powerful, cost-effective, and spatially scalable alternatives to traditional soil sampling and laboratory analysis methods (Xiao et al., 2024; Moraes et al., 2023). In this context, remote sensing enables researchers to predict terrestrial carbon content at lower costs and across more extensive geographic areas (Wang et al., 2021). Geospatial technologies facilitate data collection over large regions, thereby improving the accuracy and efficiency of SOC estimation (Wang et al., 2021). Therefore, there are different types of RS data, including multispectral, hyperspectral imagery, Radar and LiDAR data, offer distinct spectral information and spatial resolutions, each contributing uniquely to the assessment and estimation of COS (Zayani et al., 2023) See Figure 1.

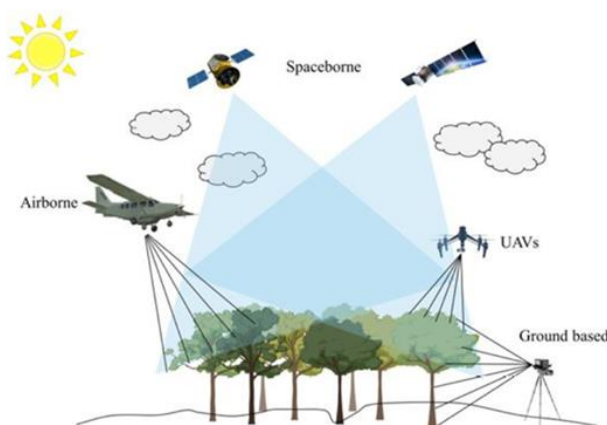


Figure 1. Remote sensing vehicles technologies, multi-platform remote sensing observation geometry (Spaceborne, airborne, UAV, and ground-based) (Tian et al., 2023)

2.1.1. Multispectral and hyperspectral imaging

Remote sensing imaging has become a valuable resource for many domains, ranging from agriculture to environmental sciences. Multispectral and Hyperspectral images are at the forefront of this technology. Spectral images are captured and analyzed based on the spectrum of light reflected, transmitted, or emitted from an object. The distinction between multispectral and hyperspectral imaging lies in the number of wavelength bands analyzed. Multispectral technology captures information across multiple discrete spectral bands that describe the reflectance properties of the Earth's surface. Satellites such as Landsat-8 and Sentinel-2 (Castaldi et al., 2019; Wang et al., 2021; Zayani et al., 2023; Tian et al., 2023; Zhou et al., 2020; Kakhani et al., 2024) provide moderate spatial resolution imagery that is useful

for mapping soil properties over large areas. These data are commonly used to predict soil organic carbon (SOC), soil texture, and other soil properties. offering broad spatial coverage and strong suitability for large-scale mapping. Zayani et al. (2023) discuss the potential of using Sentinel-2 imagery combined with laboratory data and machine learning algorithms to predict SOC.

Therefore, hyperspectral imaging provides extremely fine spectral information across hundreds of narrow wavebands, offering much higher spectral resolution than multispectral imaging. This enables more precise identification and characterization of surface materials and improves the accuracy of SOC predictions (Lin et al., 2020). The richness of hyperspectral data facilitates more accurate detection of soil attributes, thereby enhancing predictive performance. As noted by Zayani et al. (2023) and Bouslihim et al. (2025), hyperspectral data offer substantial advantages for SOC assessment, although their effectiveness may be influenced by atmospheric conditions and surface variability. These considerations apply to all satellite platforms listed in Table 1.

2.1.2. Radar and LiDAR

In recent years, Radio Detection and Ranging (Radar) and Light Detection and Ranging (LiDAR) have become essential technologies in a wide range of modern engineering applications. Both are based on a similar principle of active remote sensing (RS) of the environment, in which signals are transmitted and their echoes are analyzed, see (Table 1). LiDAR (Light Detection and Ranging) is an active remote sensing technology that measures distance by emitting laser pulses and analyzing their reflected signals. It produces highly accurate three-dimensional topographic data, can penetrate dense vegetation, and enables the generation of precise Digital Elevation Models (DEM) (He et al., 2025; Fuadah et al., 2025), whereas radar uses radio waves for object detection and terrain mapping. Furthermore, radar systems employ microwave signals to scan and capture information about the Earth's surface. Among these systems, Synthetic Aperture Radar (SAR) is a specific type of radar that is particularly valuable because it can penetrate cloud cover and acquire data both day and night, unlike optical sensors, which are dependent on weather conditions (Wang et al., 2021). Sentinel-1 is a prominent example of a satellite equipped with SAR (Wang et al., 2021; Zayani et al., 2023). SAR information can be used to estimate soil moisture, surface roughness, and other properties relevant to digital soil mapping (DSM). Zayani et al. (2023) discuss the use of SAR-derived data as covariates in SOC prediction models. Although this technology is not directly used to predict SOC, it provides valuable information on topography and vegetation structure, which can be jointly employed with other RS data to enhance and refine SOC prediction models. LiDAR, in turn, offers high-resolution elevation data, thereby providing useful indications related to SOC distribution (Kakhani et al., 2024; Fuadah et al., 2025).

Table 1. Satellite missions for spectral variables used for SOC mapping

Data Source	Number of Bands	Spatial Resolution (meters)	Temporal Resolution (days)	Radiometric Resolution (bits)	Swath (km)	Launch Year
Landsat 8	11	30	16	12	185	2013
MODIS	36	250	1-2	8	2330	1999
Sentinel 2	13	10-60	5	12	290	2015
ETM+	8	30	16	8	185	1999
PRISMA	239	30	1-2	12	30	2019
RADARSAT	1	10-100	24	6	500	1995
TerraSAR-X	1	1-100	11	8	30-100	2007
SPOT	Varies	10-30	1-3	8	60	1986
QuickBird	4	2.4	3	11	16.5	2001
WorldView1	4	0.5	1-3	8	17.6	2007
SkySat	4	2	1-3	8	15	2013

2.2. Spectral signatures and soil characteristics for SOC prediction

Remote sensing devices are capable of probing interactions between soil and electromagnetic radiation and, through these interactions, generating characteristic spectral

reflectance profiles. These profiles are subsequently used to derive various soil characteristics, among which soil organic carbon (SOC) content is of prime importance. This capability primarily stems from the fact that SOC influences soil color, which in turn affects its spectral signatures (Castaldi et al., 2019). Soil color is directly correlated with SOC content and plays an important role in soil brightness within the visible spectrum. In general, higher SOC contents tend to result in darker soils with lower reflectance in both the visible and near-infrared regions (Kakhani et al., 2024). However, the relationship between SOC and spectral reflectance is complex and is influenced by additional factors such as soil moisture, texture, and other environmental variables. Therefore, it is essential to account for these variables when developing SOC prediction models. Soil spectral characteristics, particularly those associated with SOC, are clearly expressed across several spectral regions, including the visible, near-infrared, and short wavelengths (Vaudour et al., 2022). Consequently, spectral models can be developed based on the relationships between soil reflectance and the soil property under investigation (Vaudour et al., 2022). These wavelength ranges exhibit marked sensitivity to variations in SOC content (Fu et al., 2024).

The following are some crucial soil characteristics related to soil fertility that influence SOC content, and can be assessed using remote sensing sensors:

- **Soil texture:** The proportion of fine particles: silt, clay, and sand, influences soil reflectance. Clay particles, because of their smaller size and larger surface area, tend to absorb radiation more effectively than sand. Therefore, soils with high clay content exhibit lower reflectance (Fu et al., 2024). However, this relationship is complex and can be affected by other factors, such as moisture. Fu et al. (2024) studied the influence of soil structure on SOC prediction. Similarly, Kaya et al. (2022) used sand and clay content as inputs for their predictive model.
- **Soil moisture:** Water content has a considerable impact on soil reflectance, with increased moisture causing soils to darken and reducing reflectance across the entire spectrum (Fu et al., 2024). This effect may obscure the spectral signal of SOC, underscoring the necessity of incorporating moisture effects into SOC prediction models.
- **Bulk density:** Soil bulk density is an important soil property that indicates the level of soil compaction and is often used in calculations related to organic carbon and other soil properties, making it an important consideration in predicting SOC stocks (Zaher et al., 2019; Bahri et al., 2022; Chen et al., 2024).
- **Soil mineralogy:** Soil minerals have unique characteristic spectra that exhibit specific absorption features, allowing the use of spectral signatures to differentiate soil types. Identifying these minerals can refine SOC predictions to some extent and contribute to soil classification. However, differences in mineral composition lead to variations in soil color and, consequently, in SOC content (Castaldi et al., 2019; Fu et al., 2024).
- **Other characteristics:** SOC content depends on various soil properties, such as electrical conductivity, pH, and drainage capacity. Electrical conductivity and pH are of paramount importance because they influence microbial activity and, consequently, SOC composition, making them important covariates in most predictive models. The laboratory determination of electrical conductivity and pH for modeling purposes was reported by Zhang et al. (2023).

2.3. Environmental covariates

Climatic parameters such as temperature and precipitation, along with topography, vegetation cover, LCLU, and spectral indices, contribute to increased accuracy and reliability in SOC predictions. When combined with environmental variables derived from remote sensing (RS), these factors significantly enhance predictive models by providing substantial information on the drivers of SOC distribution. All the aforementioned variables influence

SOC decomposition and vegetation productivity, both of which directly or indirectly affect SOC accumulation and its spatial distribution within the study area.

2.3.1. Spectral indices

Spectral indices (SIs) are a general term for mathematical equations applied to raster image pixels to characterize land-based phenomena. They are usually derived from reflectance data obtained after preprocessing multispectral remote sensing (RS) images. These indices are essential for soil organic carbon (SOC) prediction because they link soil attributes with spectral reflectance (Zhou et al., 2020). Most SIs are derived from remote sensing imagery, and numerous studies have employed different multispectral optical sensors, such as Landsat, MODIS, and SPOT (Zhou et al., 2020; Vaudour et al., 2022). Several widely used SIs, including NDVI, EVI, and SATVI, show strong correlations with SOC (Zhou et al., 2020). More recently, SOC prediction has also been enhanced by incorporating NDVI and GNDVI alongside other environmental variables (Kaya et al., 2022). According to Wang et al. (2021), although vegetation indices are important, their inclusion in models that also rely on terrain variables provides even greater predictive power. However, the transferability and applicability of these indices may depend on soil type, mineralogy, and moisture conditions (Castaldi et al., 2019), Table 2.

Table 2. Spectral indices used for SOC prediction in some studies

Index	Definition	Reference
Normalized Difference Vegetation Index (NDVI)	$NDVI = \frac{NIR - Red}{NIR + Red}$	Kakhani et al. (2019) ; Dong et al. (2024)
Soil-Adjusted Vegetation Index (SAVI)	$SAVI = \frac{(NIR - Red)}{(NIR + Red + L)} (1 + L)$	Wang et al. (2021)
Enhanced Vegetation Index (EVI)	$EVI = \frac{G(NIR - Red)}{NIR + C_1 Red - C_2 Blue + L}$	Dong et al. (2024) ; Cutting et al. (2024)
Land Surface Water Index (LSWI)	$LSWI = \frac{NIR - SWIR}{NIR + SWIR}$	Zayani et al. (2023) ; Emadi et al. (2020)
Brightness Index (BI)	$BI = \sqrt{(Red^2 + Green^2)}$	Wang et al. (2021); Fathizad et al. (2022)
Bare Soil Index (BSI)	$BSI = \frac{(SWIR + Red) - (NIR + Blue)}{(SWIR + Red) + (NIR + Blue)}$	Zayani et al. (2023)
Clay Index (CI)	$CI = \frac{SWIR1}{SWIR2}$	Wang et al. (2021) ; Emadi et al. (2020)
Ratio Vegetation Index (RVI)	$RVI = \frac{NIR}{Red}$	Wang et al. (2021)
Normalized Difference Water Index (NDWI)	$NDWI = \frac{Green - NIR}{Green + NIR}$	Dong et al. (2024)
Difference Vegetation Index (DVI)	$DVI = NIR - Red$	Wang et al. (2021)
Soil-Adjusted Total Vegetation Index (SATVI)	$SATVI = \frac{(SWIR - Red)}{(SWIR + Red + 1)} * 2 - SWIR/2$	Zhou et al. (2021)
Modified Soil-Adjusted Vegetation Index (MSAVI)	$MSAVI = \frac{2NIR + 1 - \sqrt{(2NIR + 1)^2 - 8(NIR - Red)}}{2}$	Cutting et al. (2024)

2.3.2. Land cover and land use

Land cover refers to the physical characteristics observed on the Earth's surface, such as different types of vegetation, whether natural or cultivated. In contrast, land use describes human interactions with a specific type of land cover, including the activities, developments, and resource use involved in modifying, producing, or preserving it. Land cover and land use (LCLU) maps derived from remote sensing data are widely used to classify vegetation and land use types. The importance of land use in soil organic carbon (SOC) formation varies depending on vegetation type; therefore, LCLU variables are important covariates in SOC modeling. For example, Zeraatpisheh et al. (2023) included land cover as a categorical variable in their SOC prediction models. In other studies, Wang et al. (2021) evaluated SOC prediction under different land-use scenarios and demonstrated significant differences in

prediction accuracy. Sabir et al. (2020) argued that anthropogenic effects, particularly in land-use types such as agriculture and forestry, play a crucial role in SOC storage. Moreover, the historical characteristics of agriculture and land use are also important for understanding SOC accumulation over time.

2.3.3. Climate covariates

Climate indicator variables represent specific factors that define the state of the climate system at a given time and location. Precipitation and temperature are examples of such variables and can be obtained from remote sensing platforms or readily retrieved from meteorological databases. These variables significantly influence soil organic carbon (SOC) levels. Temperature primarily controls decomposition rates, whereas precipitation affects soil moisture and biomass production. In this study, Zeraatpisheh et al. (2023) identified temperature and precipitation as the most critical predictors of SOC. Similarly, Kaya et al. (2022) incorporated climate variables into their models, and Xiao et al. (2024) demonstrated that precipitation plays a key role in improving the accuracy of SOC prediction.

2.3.4. Topographic attributes

The terrain plays a significant role in the spatial distribution of soil organic carbon (SOC), with its effects varying across different environmental variable classes (Xiao et al., 2024). When using terrain features derived from digital elevation models (DEMs) in SOC prediction models, it is essential to extract landscape characteristics that influence SOC distribution. Key terrain attributes obtained from DEMs serve as important covariates (Zhang et al., 2023). Water movement and distribution, in particular, affect soil development and SOC accumulation. Several studies, including those by Kaya et al. (2022), Zeraatpisheh et al. (2023), and Khosravi et al. (2021), confirm the role of topographic attributes, including DEM derivatives, in predicting SOC content. All of the attributes used are extracted from DEMs, as summarized in Table 3.

Table 3. Global open DEM data for topography covariates

DEM	Source	Spatial Resolution (meters)	Vertical Accuracy (meters)	Reference
ASTER GDEM	NASA METI	30	10 - 25	Xiao et al. (2024)
NASADEM	NASA	30	3 - 5	Morais et al. (2023)
Copernicus DEM	ESA	30	4	Zhou et al. (2020)
MERIT DEM	MERIT	90	1.9 - 3.5	Dong et al. (2025)
SRTM DEM	NASA	30	5 - 10	Taghizadeh et al. (2020)

3. Machine Learning Methods for SOC Prediction

3.1. Machine learning (ML)

Machine learning is an integral field of artificial intelligence. It defined learning as a process by which a model improves its performance relying on data, sometimes hidden, to discover underlying rules, laws, or logic behind complex phenomena. In this way, ML enables the automatic extraction of crucial information from data through sophisticated analytical techniques (Ahansal et al., 2022). Recently, ML has grown rapidly, with applications spanning medicine, industry, agriculture, and geomatics. These applications come in diverse forms, each representing a wide variety of potential models and algorithmic approaches.

Machine learning (ML) is a robust tool and a valuable supplement for soil organic carbon (SOC) prediction. It is both economically efficient and more effective than conventional methods. Table 4 summarizes the advantages and limitations of each algorithm. ML algorithms can reveal highly complex relationships between SOC and various environmental variables and soil characteristics. These relationships enhance the predictive accuracy of SOC content at unsampled locations, enabling the development of SOC maps for large-scale assessments. This facilitates a better understanding of the spatial variability of SOC and supports decision-making for sustainable soil management. Heuvelink et al. (2020) described a data-driven ML model calibrated using SOC field observations and incorporating both static and dynamic covariates. Several studies, including those by Emadi et al. (2020),

Heuvelink et al. (2020), Cheng et al. (2024), Zhou et al. (2019), and Rostaminia et al. (2021), have evaluated the performance of various ML methods such as random forest (RF), artificial neural networks (ANN), deep neural networks (DNN), Cubist (CB), support vector machines (SVM), gradient boosting machines (GBM), and XGBoost for SOC prediction. These models provide precise estimates of SOC levels and are highly beneficial for farmers and agricultural managers, enabling informed decision-making related to sustainable practices, soil conservation, and carbon sequestration strategies (Mango et al., 2024).

3.2. Support vector machines (SVM)

Support vector machines (SVMs) are algorithms designed for classification and regression tasks (Emadi et al., 2020; Cheng et al., 2024). Although initially developed for classification, they were later extended to address regression problems. SVM algorithms are known for their ability to accurately approximate multivariable functions with a high degree of precision (Emadi et al., 2020). They are particularly efficient in handling high-dimensional data and nonlinear relationships, making them well suited for predicting soil organic carbon (SOC) using multiple environmental variables (Cheng et al., 2024).

3.3. Random forest (RF)

Random forest (RF) is a supervised learning algorithm valued for its versatility and ease of use (Manteghi et al., 2024). It is widely used due to its ability to handle complex non-linear relationships and its robustness against overfitting. RF is an ensemble algorithm that combines the outputs of multiple decision trees to generate predictions for both classification and regression tasks (Zhou et al., 2019). During the training phase, the algorithm constructs multiple decision trees and then combines their results to make predictions (Zhou et al., 2019). Each tree is trained on a bootstrap sample of the data selected with replacement, and the algorithm chooses the best data subsets for model training (Manteghi et al., 2024).

3.4. Artificial neural networks (ANN)

Artificial neural networks (ANNs) are distinguished from other models by their use of computational methods inspired by the neural functions of the human brain (Emadi et al., 2020). As summarized in Table 4, these models have several strengths and weaknesses. ANNs are capable of capturing complex nonlinear relationships within large datasets, making them potentially useful for soil organic carbon (SOC) prediction (Cheng et al., 2024; Espinel et al., 2024). However, they often require large amounts of training data and can be difficult to interpret (Mishra et al., 2020). Furthermore, ANNs are widely used in agricultural mapping, particularly for soil moisture estimation and resource management applications (Espinel et al., 2024).

3.5. Cubist

Similar to the Random forest (RF) approach, the Cubist algorithm is a data-driven tree-based regression method used to examine nonlinear relationships between a target variable and predictor variables. It generates multivariate models by combining rule-based trees with regression algorithms (Rostaminia et al., 2021). The Cubist model integrates multiple linear regression sub-models, allowing it to effectively handle nonlinear relationships while offering improved interpretability compared with other machine learning algorithms (Ou et al., 2024).

3.6. Gradient boosting machines (GBM)

Gradient boosting machine (GBM) is an algorithm that constructs decision trees sequentially, where each new tree focuses on correcting the residuals of the previous ones. The algorithm combines several “weak learners” to form a “strong learner” (Mishra et al., 2020). It is based on the principle of boosting, in which models are generated iteratively to rectify errors made by earlier models (Mishra et al., 2020). Each observation is assigned a

weight according to its prediction performance, giving greater emphasis to poorly predicted instances when constructing subsequent models (Meliho et al., 2023).

3.7. Extreme gradient boosting (XGBoost)

Extreme gradient boosting (XGBoost) is a widely used and powerful method that represents an improved implementation of gradient boosting algorithms, proven to be faster and to achieve better performance (Zhang et al., 2022). By combining multiple weak learners to form a strong learner, XGBoost is designed to perform computations efficiently while reducing the risk of overfitting (Espinell et al., 2024). It is based on the gradient boosting framework but incorporates enhancements such as regularization to control model complexity and parallelized computation to improve training speed (Zhang et al., 2022).

3.8. Deep neural network (DNN)

Deep neural networks (DNNs) are a type of artificial neural network composed of multiple layers of interconnected neurons that function in a manner similar to the human brain. These networks differ from other neural network models because of their architecture, which includes multiple hidden layers between the input and output layers. The depth of these layers enables the network to learn complex nonlinear relationships in data through multiple levels of abstraction (Zayani et al., 2023; Liakos et al., 2018). Due to their ability to model complex interactions, DNN are particularly powerful tools in various fields, including agriculture, where they offer high accuracy and can automatically learn relevant features from data. They are commonly used as reliable and efficient approximators to analyze relationships between input and output variables, whether linear or nonlinear (Emadi et al., 2020).

Table 4. Summary of some strengths and weaknesses of ML algorithms cited

ML Algorithm	Strengths	Weaknesses
Random Forest (RF) (Adeniyi et al., 2024; John et al., 2020)	Effectively handles high-dimensional data, achieves high accuracy, reduces overfitting compared with decision trees, captures complex interactions between variables.	Computationally expensive for large datasets; the process can be slower due to the need to evaluate many trees and is less interpretable than a single tree.
Artificial Neural Networks (ANN) (Emadi et al., 2020; John et al., 2020)	Can model highly complex relationships; works well with large and unstructured datasets; learns hierarchical representations; and is robust to noisy data.	Requires a large dataset for training, is computationally intensive, is difficult to interpret, and requires large amounts of memory and computational resources.
Cubist (CB) (Zhu et al., 2022; Mahmoudzadeh et al., 2020)	High accuracy, handles high-dimensional data, is robust to missing data and noise, is faster than complex tree ensembles, and can model non-linear relationships by dividing the data into regions and fitting a linear model within each region.	Less flexible than ensembles, works best with moderate to large datasets, requires careful preprocessing, can become complex with many rules and trees, is harder to interpret and is prone to overfitting.
Gradient Boosting Machines (GBM) (Zhang et al., 2023; Meliho et al., 2023)	High predictive accuracy, handles missing and noisy data, is less prone to overfitting with proper tuning, captures nonlinear relationships, is flexible, robust to overfitting, and works well with unstructured data.	Requires careful hyperparameter tuning; training can be slow, especially on large datasets; is harder to interpret than simple regression models; and struggles with very small datasets.
Extreme Gradient Boosting (XGBoost) (Kumar et al., 2023; Tan et al., 2022).	Handles high-dimensional data, achieves high predictive accuracy, is efficient on large datasets, and handles missing data.	Struggles with very small datasets, requires careful parameter tuning, and has limited interpretability compared to simpler models.
Support Vector Machines (SVM) (Ding et al., 2023; Guo et al., 2025).	Effective in high-dimensional spaces, suitable for small to medium datasets, can handle noisy data, works well with small datasets, and is robust to overfitting with proper tuning	Computationally expensive for large datasets, requires careful kernel selection, struggles with highly imbalanced data, and requires feature scaling.
Deep Neural Network (DNN) (Taghizadeh et al., 2020; Liakos et al., 2018).	Adaptability to large datasets, scalability, ability to handle multidimensional data, and modeling of complex relationships.	Computational cost, overfitting, interpretability issues, data-intensive requirements, and sensitivity to data quality.

4. Remote Sensing, GIS, and Machine Learning Approach for SOC Prediction

4.1. SOC Prediction using RS-GIS-ML Approaches

Integrating remote sensing (RS), geographic information system (GIS) data, and machine learning (ML) algorithms as data processing and modeling tools provides an effective method for predicting soil organic carbon (SOC) levels across both small and large areas. RS data offer a wealth of information, including spectral indices, land cover and land use, and topographic features, all of which are related to SOC content. These features are then used to train ML models capable of capturing complex and nonlinear relationships between RS–GIS data and SOC content. The resulting models can predict SOC levels across the study region, enabling scalable and non-invasive soil health monitoring. This approach provides essential information for environmental monitoring without the need for intensive soil sampling. Accordingly, Figure 2 summarizes the SOC prediction process based on the studied approaches and illustrates the main steps, which typically involve the following key stages:

- **Data acquisition:** This step involves collecting soil samples in the field for SOC measurement and integrating their geographic coordinates with environmental data relevant to SOC distribution. The soil samples are then analyzed in the laboratory to determine SOC content. Environmental data may include remote sensing data, which provide spectral information about the Earth's surface in the study area. Additionally, other geospatial datasets such as topographic attributes derived from digital elevation model (DEM) and climate data (e.g., temperature and precipitation) obtained from meteorological sensors are often incorporated to improve prediction accuracy (Zhou et al., 2020; Fathizad et al., 2022; Sodango et al., 2021).
- **Feature Engineering:** Feature engineering is crucial for improving the accuracy of SOC predictions by preprocessing data and transforming raw inputs into meaningful variables through appropriate procedures, thereby optimizing model performance. Remote sensing data provide multispectral and hyperspectral information, facilitating the extraction of spectral indices. Additionally, topographic variables can be derived from digital elevation model (DEM) (Emadi et al., 2020). Beyond the creation of new features such as land cover and land use maps, data transformation and feature selection techniques, including standardization and dimensionality reduction, are applied to ensure that only the most relevant variables are used for SOC prediction.
- **Machine learning model training:** Various machine learning algorithms are employed to model the relationships between extracted features and SOC content measured from soil samples. These algorithms are trained to learn the relationship between predictor variables (remotely sensed and environmental data) and the target variable (SOC content). Commonly used algorithms include Random Forest (Chinilin et al., 2023), Support Vector Machines, and Gradient Boosting (Kumar et al., 2023), each with its own strengths and weaknesses. Furthermore, the ML models are trained using datasets from locations where SOC measurements are available.
- **Model evaluation and validation:** Validation and testing of the trained ML models are conducted using unseen datasets to assess their generalization ability, validity, and predictive accuracy. Commonly used evaluation metrics include the coefficient of determination (R^2), root mean square error (RMSE), mean absolute error (MAE), and concordance correlation coefficient (CCC), among others (Emadi et al., 2020; Meliho et al., 2023; Xie et al., 2022; Mahmood et al., 2024). Based on the validation results, the trained models can be further improved through strategies such as

hyperparameter tuning, incorporating additional data, or testing alternative ML algorithms.

- **SOC prediction and mapping:** The finalized model is then used to generate a geographic representation of SOC distribution across the surveyed area based on features derived from remote sensing (RS), GIS, and environmental variables. It predicts SOC content and provides continuous geospatial information on its spatial variability, enabling the identification of zones with increasing or decreasing carbon levels. This spatial insight forms a critical foundation for soil conservation strategies, environmental monitoring, and various land management applications (Zhou et al., 2020; Xie et al., 2022).

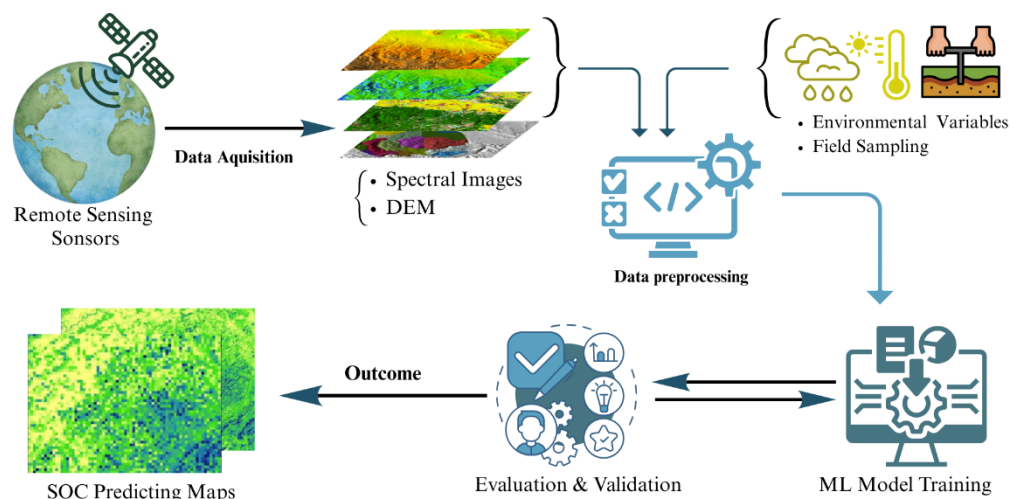


Figure 2. Workflow for soil organic carbon (SOC) prediction using remote sensing, GIS and machine learning

4.2. Studies of SOC prediction on global levels

Owing to the increasing demand for accurate soil maps and SOC distribution estimates, along with the growing ease of access to remote sensing (RS) imagery, climate data, and digital elevation model (DEM), machine learning techniques have been increasingly adopted for SOC prediction (Emadi et al., 2020). With rapid advancements in RS and information technology, digital soil mapping (DSM) has emerged as a rapidly growing discipline. ML algorithms can effectively identify complex relationships between soil characteristics and predictor variables, thereby supporting the prediction of soil characteristics in areas with limited soil information (Zhou et al., 2020). Consequently, tree-based ensemble ML methods, such as Random Forests, Cubist, and Boosted Regression Trees, have become standard approaches for SOC prediction because of their ability to model complex nonlinear relationships and efficiently handle high-dimensional datasets (Zhou et al., 2020; Emadi et al., 2020), as shown in **Table 5** and

Table 6. In addition to ensemble techniques, other algorithms such as deep neural networks (DNN), artificial neural networks (ANN), and support vector machines (SVM) also offer significant advantages for soil property mapping due to their diverse capabilities in capturing and modeling complex data relationships.

Several studies have demonstrated that integrating remote sensing (RS) data with machine learning (ML) algorithms is effective for SOC mapping. In this context, Emadi et al. (2020) conducted a study in northern Iran in which several ML models, including deep neural networks, were applied in combination with RS data to predict SOC content. The results showed high predictive accuracy, particularly for the MIT-based model. Kumar et al. (2023) also carried out a study in the Upper Brahmaputra Valley in Northeast India using Random

Forest, Cubist, Extreme Gradient Boosting, and Support Vector Machines, and found that the Random Forest model produced the most accurate SOC estimates. In another study, Chinilin et al. (2023) applied a Random Forest based ML approach to three-dimensional modeling of SOC distribution across soil layers in Russia. Additionally, Kumar et al. (2023) investigated the use of Landsat 8 imagery combined with ML models, mainly XGBoost and Random Forest, for digital SOC mapping in northern Iraq. Their results highlighted the importance of spectral imagery and topographic variables in improving the accuracy of SOC prediction models.

Numerous studies have been conducted to evaluate the performance of various ML algorithms for SOC prediction, including Random Forest (RF), Support Vector Machines (SVM), XGBoost, deep neural networks (DNN), Cubist, and gradient boosting machines (GBM). In this context, studies addressing nonlinear relationships and complex interactions between SOC and environmental predictors have often reported superior performance for RF and XGBoost models. For example, in northern Iran, Emadi et al. (2020) found that DNNs provided the highest prediction accuracy, with precipitation identified as the most influential variable. Conversely, Khalaf et al. (2023) reported that in northern Iraq, XGBoost outperformed RF in soil organic matter (SOM) prediction. On the other hand, Kumar et al. (2023) indicated that for SOC mapping in India, RF was the most effective model, whereas Cubist showed the lowest performance. In contrast, Meliho et al. (2023) documented that in the Moroccan High Atlas region, both RF and Cubist achieved the highest accuracy in predicting SOC stocks.

To date, most significant studies have demonstrated that the predictive performance of machine learning algorithms such as Random Forest, XGBoost, and deep neural networks, can be very high, largely due to the inclusion of key predictors such as climatic variables, topographic features, and spectral indices, which strongly influence SOC variability. Common metrics used to assess model performance include the coefficient of determination (R^2), root mean square error (RMSE), mean absolute error (MAE), and the concordance correlation coefficient (CCC). Overall, these studies demonstrate or strongly indicate the considerable potential of ML-based approaches to produce accurate and efficient SOC predictions and spatial maps. These advances also contribute to the development of informed strategies for climate change mitigation, as they are critical for applications such as sustainable soil management and carbon budget assessment.

Table 5. Examples of studies addressing the use of ML algorithms for SOC prediction

Country	Evaluation metrics	Area (km ²)	Environmental Covariates				Reference
			Number	Topography	Climate	Spectral	
France	RPD, RPIQ, R^2 , RMSE	1.5	26	✓		✓	Zayani et al. (2023)
China	MAE, RMSE, R^2	5475.5	12	✓	✓	✓	Zhang et al. (2023)
Tunisia	RMSE, ME, R^2	163,610	36	✓	✓	✓	Bahri et al. (2022)
China	R^2 , RMSE, RPIQ	33920	17	✓	✓	✓	Dong et al. (2024)
Iran	MAE, R^2 , RMSE, CCC	23881	105	✓	✓	✓	Emadi et al. (2020)
Argentina	ME, RMSE, CCC	2780000	35	✓	✓	✓	Heuvelink et al. (2020)
Morocco	MAE, ME, R^2 , RMSE	-----	88	✓	✓	✓	Meliho et al. (2023)
China	RMSE, R2	377900	29	✓	✓	✓	Zhang et al. 82022)
Italy	CCC, RI, RMSE	314	8	✓			Adeniyi et al. (2024)
Nigeria	MAE, R^2 , RMSE	60	21	✓		✓	John et al., 2020)
Iran	MAE, RMSE, R^2	29043	101	✓	✓	✓	Mahmoudzadeh et al. (2020)
India	RMSE, R^2 , ME, CCC	2277	28	✓	✓	✓	Kumar et al., 2023)
China	MAE, RMSE, R^2	2667	55	✓	✓	✓	Tan et al. (2022)
Iraq	MAE, RMSE, R^2	528	13	✓		✓	Khalaf et al. (2023)
China	R^2 , MAE, MSE	7567	10	✓	✓	✓	Guo et al. (2025)
China	RSME, R^2	11462.41	52	✓	✓	✓	Sodango et al., 2021)
Russia	RMSE, R^2 , CCC	17100000	22	✓	✓	✓	Chinilin et al. (2023)
Germany	R^2 , RMSE, AIC, MAPE, BIC, MAE	357022	24	✓	✓	✓	Sakhaee et al. (2022)
Türkiye	ME, MAE, RSME	1330	10	✓	✓	✓	Kılıç, et al. (2022)
China	MSE, MAE, R^2 , RSME	10624	12	✓	✓	✓	Li et al. (2025)

Table 6. Summarizing key studies with samples, metrics, and best-performing models

Study Reference	Soil Samples	Selected Algorithm	Performance Metrics	Interpretation
Zayani et al. (2023)	58	<i>DNN</i>	RPD= 4.51 RPIQ= 3.90 R ² = 0.95 RMSE= 0.85 g·kg ⁻¹	Very high predictive accuracy. Strong nonlinear relationships between spectral indices and SOC. Laboratory + Sentinel-2 indices improved performance. Indicates SOC variability was strongly controlled by spectral reflectance patterns.
Zhang et al. (2023)	117	<i>BPNN</i>	MRE= 0.066 RMSE= 0.257 g/kg	Extremely low prediction error. Agricultural soils with structured patterns. Neural network captured spatial trends effectively.
Bahri et al. (2022)	1540	<i>QRF</i>	RMSE= 1.94 kg m ⁻² ME= 0.19 kg m ⁻² R ² = 0.44	Moderate predictive ability. National-scale mapping increases heterogeneity. QRF useful for uncertainty quantification.
Dong et al. (2024)	73	<i>CNN-LSTM</i>	R ² = 0.69 RMSE= 6.06 g kg ⁻¹ RPIQ= 1.96	Combines spatial (CNN) + temporal (LSTM) patterns. Effective in complex vegetation environments. Better than RF and standalone CNN/LSTM.
Emadi et al. (2020)	1879	<i>DNN</i>	MAE= 0.59 % R ² = 0.65 RMSE= 0.75 % CCC= 0.83	Good agreement between predicted and observed SOC. Complex mountainous terrain required nonlinear modeling.
Heuvelink et al. (2020)	5114	<i>QRF</i>	ME= 0.03 kg C m ⁻² RMSE= 2.04 kg C m ⁻² CCC= 0.631	Focus on SOC change over time. Emphasis on uncertainty, not just accuracy.
Meliho et al. (2023)	420	<i>RF</i>	MAE= 5.33 t/ha ME= -0.07 t/ha R ² = 0.92 RMSE= 7.58 t/ha	Very high accuracy. Strong relationship between terrain + climate + SOC stocks. RF stable and robust.
Zhang et al. (2022)	1584	<i>RF</i>	RMSE= 2.80–3.49 g kg ⁻¹ R ² = 0.58–0.74	Performance improved with depth. RF handled multicollinearity well.
Adeniyi et al. (2024)	120	<i>RF-RK</i>	CCC= 0.38 RI= 0.81 % RMSE= 0.35 g kg ⁻¹	Residual kriging improved spatial structure capture. Combines machine Learning + geostatistics.
John et al. (2020)	60	<i>RF</i>	MAE= 0.17 R ² = 0.68 RMSE= 0.20	Acceptable accuracy. Alluvial terrain relatively structured.
Mahmoudzadeh et al. (2020)	865	<i>RF</i>	MAE= 0.30% RMSE= 0.35% R ² = 0.60	Moderate accuracy. Semi-arid conditions increase variability.
Kumar et al. (2023)	160	<i>RF</i>	RMSE= 0.377 % R ² = 0.418 ME= 0.136 CCC= 0.549	Calibration high but validation moderate. Indicates potential overfitting in other models. RF more generalizable.
Tan et al. (2022)	186	<i>RF</i>	MAE= 2.79 g kg ⁻¹ RMSE= 3.68 g kg ⁻¹ R ² = 0.71 RPD = 1.86	Multisource covariates improved performance. Scale strongly influences accuracy.
Khalaf et al. (2023)	96	<i>XGBoost</i>	MAE= 0.41 RMSE= 0.62 R ² = 0.92	Excellent predictive performance. Gradient boosting captured complex interactions.
Guo et al. (2025)	11382	<i>RF</i>	R ² = 0.81 MSE = 0.41 kg/m ² MAE = 0.34 kg/m ²	The Random Forest model explains 81 % of the variability in soil organic carbon (SOC). The low MSE and MAE values indicate small prediction errors.

Table 7. Cont.

Study Reference	Soil Samples	Selected Algorithm	Performance Metrics	Interpretation
Sodango et al. (2021)	121	RF	$R^2 = 0.96$ RMSE = $0.91 \text{ mg} \cdot \text{g}^{-1}$	The very high R^2 indicates excellent predictive performance. The low RMSE shows high accuracy and small deviations between predicted and observed SOC values.
Chinilin et al. (2023)	863	RF	$R^2 = 0.45$ CCC = 0.63 RMSE = 1.41 g/kg	The performance is moderate when spatial cross-validation is applied. The decrease in R^2 compared to random validation suggests that spatial autocorrelation can artificially inflate model performance.
Sakhaee et al. (2022)	3501	SVM	RMSE = 21.3 g kg^{-1} MAE = 8.7 g kg^{-1} MAPE = 34.3 %	Separating mineral and organic soils significantly improved prediction accuracy. SVR performs well with limited sample sizes and captures nonlinear relationships using kernel functions.
Kilic, et al. (2022)	132	ANN-OK	ME = -0.028 MAE = 0.042 RMSE = 0.066	The success of the ANN-OK model demonstrates that combining machine learning with geostatistics provides a more comprehensive representation of soil organic carbon variability.
Li et al. (2025)	512	RF	MSE= 0.0312 MAE= 0.1244 $R^2= 0.4327$ RMSE= 0.1768	The Stacking model combines RF, AdaBoost, and SVR with a Ridge regression meta-learner. It achieves the best balance between model fit and generalization. It has the lowest MAE, MSE, and RMSE among the tested models.

CNN-LSTM: Convolutional Neural Network- Long Short-Term Memory, **OK:** Ordinary Kriging, **BPNN:** Back-Propagation Neural Network, **QRF:** Quantile Regression Forest, **RK:** Residual Kriging

4.3. Evaluation and validation of prediction models

Evaluation and validation are among the most important steps in ensuring the reliability and accuracy of soil organic carbon (SOC) prediction models. These methods are used to assess how well a model performs and whether its predictions are representative of reality. Evaluation refers to the assessment of a model's performance using specified metrics, allowing an understanding of how well the model represents the underlying SOC distribution. Validation, on the other hand, involves testing the model using an independent dataset that does not take part in the training process. This ensures that the model's

predictive accuracy is not biased by overfitting or by characteristics specific to the training dataset, see Figure 3.

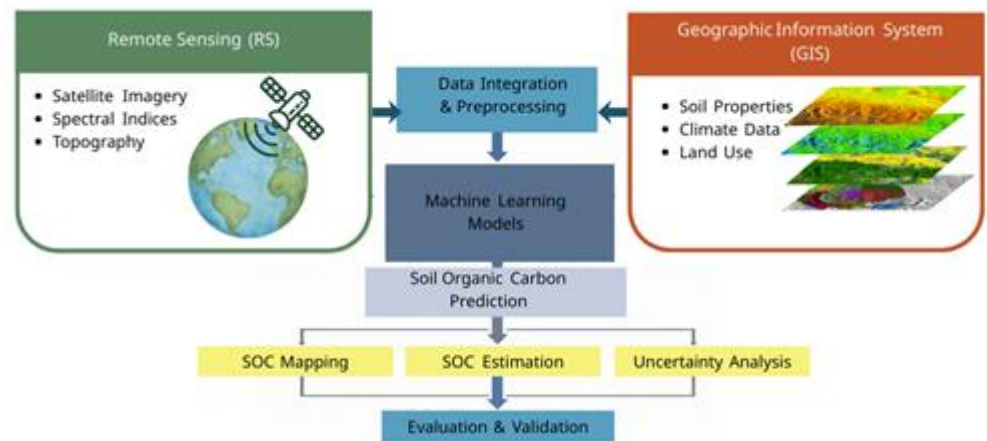


Figure 3. Integrated remote sensing-GIS-machine learning framework for soil organic carbon prediction, mapping, and uncertainty analysis

4.3.1. Commons evaluation metrics

Coefficient of Determination (R^2): A numerical metric used to measure the difference portion of the observed state-of-charge (SOC) data that becomes actually reflected by prediction model; the higher the rate (i.e. gets closer to 1), indicates a better model fit (Zhang et al., 2022). However, having a high R^2 doesn't mean the model is perfect but affected by overfitting. This issue tends to happen with complex model or noisy data (Emadi et al., 2020; Ding et al., 2023) (Equation 1).

$$R^2 = 1 - \frac{\sum_{i=1}^n (p_i - o_i)^2}{\sum_{i=1}^n (\hat{p}_i - \hat{o}_i)^2} \quad (1)$$

Root Mean Square Error (RMSE): Another widely used metric for assessing the performance of predictive models, including those for SOC prediction. RMSE measures the average discrepancy between predicted and observed SOC values, thereby indicating how much the model's predictions differ from the actual observed values. Lower RMSE values indicate higher accuracy (Zhang et al., 2022). Because RMSE assigns greater weight to larger errors, it is particularly sensitive to large deviations. Additionally, RMSE is expressed in the same units as the SOC data, making it easy to interpret. Along with R^2 , RMSE is commonly used to compare predictive methods (Emadi et al., 2020; Ding et al., 2023). See Table 8 for more metrics (Equation 2).

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (p_i - o_i)^2} \quad (2)$$

Table 8. Others evaluation metrics used in selected studies

Metrics	Equation	Reference
Ratio of Performance to Interquartile Distance (RPIQ)	$RPIQ = \frac{Q3 - Q1}{RSME}$	Zayani et al. (2023); Xie et al. (2022)
Ratio of Performance to Deviation (RPD)	$RPD = \frac{STD}{RSME'}$	Zayani et al. (2023)
Mean Absolute Percentage Error (MAPE)	$MAPE = \frac{1}{n} \sum_{i=1}^n \left \frac{y_i - \hat{y}_i}{y_i} \right \times 100$	Sakhaee et al. (2022)

Akaike Information Criterion (AIC)	$AIC = 2k - 2 \ln(L)$	Sakhaee et al. (2022)
Bayesian Information Criterion (BIC)	$BIC = \ln(n) k - 2 \ln(L)$	Sakhaee et al. (2022)
Relative Improvement (RI)	$RI = \frac{RMSE_{baseline} - RMSE_{model}}{RMSE_{baseline}} \times 100$	Adeniyi et al. (2024)
Mean Square Error (MSE)	$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$	Li et al. (2025)
Mean Error (ME)	$ME = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)$	Kumar et al. (2023) ; Sodango et al. (2021)
Mean Absolute Error (MAE)	$MAE = \frac{1}{n} \sum_{i=1}^n (p_i - o_i)$	Patriche et al. (2023) ; Nosari et al. (2024)
Concordance Correlation Coefficient (CCC)	$CCC = \frac{2\rho\sigma_o\sigma_p}{\sigma_o^2 + \sigma_p^2 + (\mu_o - \mu_p)^2}$	Emadi et al. (2020) ; Adeniyi et al. (2024)

4.3.2. Uncertainty assessment

Uncertainty assessment is very important when predicting soil organic carbon, as its distribution depends on a complex system of interacting factors. It is therefore essential to provide measures of the uncertainty inherent in SOC predictions in order to understand the level of confidence and the extent of uncertainty in the resulting predicted maps.

Methods for uncertainty assessment:

Bootstrapping: A non-parametric method used to measure spatially related prediction uncertainty by generating multiple realizations of the dataset through resampling and using these realizations to produce predictions (Kumar et al., 2023).

Confidence Intervals: These are intervals estimates that define the range in which the actual SOC value may fall. For example, a 90 % confidence interval indicates that the respective value is contained in the calculated interval with a probability of 90% (Emadi et al., 2020; Moharana et al., 2024).

Prediction Interval Coverage Probability: A metric used to evaluate how well the predicted uncertainty performs in practice. Ideally, the PICP should be as close as possible to the nominal confidence level of the prediction intervals (Kumar et al., 2023; Moharana et al., 2024).

Evaluation and uncertainty assessment provide essential checks on the performance of SOC prediction methods to ensure that resulting maps are reliable and accurate. High levels of confidence in model predictions, supported by appropriate evaluation and validation, allow these models to serve as robust tools for land management decision-making, climate change modeling, and agricultural action planning. Without proper evaluation and validation, SOC model predictions may be inaccurate or misleading, thereby limiting their practical usefulness for achieving sustainable soil management and effective carbon sequestration strategies.

4.3.3. Factors influencing uncertainty in soc prediction

Several factors contribute to uncertainty in predicting soil organic carbon (SOC), with sampling density being one of the most influential. Uncertainty is generally higher in areas with fewer sampling points, as this limits the model’s ability to accurately represent spatial variations in SOC levels (Zhang et al., 2022). In addition to sampling distribution, the quality and quantity of environmental variables incorporated into the prediction model have a substantial impact on prediction uncertainty. Insufficient or low-quality environmental data can lead to unreliable model outputs (Moharana et al., 2024). Furthermore, limitations

inherent to the models themselves, including uncertainties in the relationships between SOC and environmental variables, as well as model-related errors can further increase prediction uncertainty and complicate SOC estimation (Emadi et al., 2020). Collectively, these factors highlight the complexity of SOC prediction and the challenges researchers face in minimizing uncertainty.

5. Discussion

Geospatial technologies and machine learning (ML) have profoundly changed digital soil mapping (DSM), particularly for soil organic carbon (SOC) prediction. Beyond providing cost-effective and spatially extensive data, these approaches allow the exploration of complex nonlinear relationships between SOC and environmental covariates. However, the relevance of these results depends strongly on the nature of the input data and on the modeling strategy adopted. Remote sensing data, especially visible, near-infrared (NIR) and shortwave infrared (SWIR) bands, are widely used because they are readily available and sensitive to surface soil and vegetation properties. Spectral indices such as the Normalized Difference Vegetation Index (NDVI), Soil-Adjusted Vegetation Index (SAVI), Brightness Index (BI), and Enhanced Vegetation Index (EVI) have shown reasonable performance for SOC estimation, particularly in arid and semi-arid environments where vegetation cover is sparse (Emadi et al., 2020; Gunal et al., 2024). Nevertheless, these indices mainly capture surface conditions and vegetation status, which limits their ability to represent subsurface SOC dynamics and makes them sensitive to confounding factors such as soil moisture, roughness and land cover.

Machine learning models such as Random Forest (RF), Deep Neural Networks (DNN), Cubist and XGBoost have demonstrated strong predictive power in SOC mapping (Ding et al., 2023; Wang et al., 2024). RF and XGBoost are generally more robust to noisy and collinear predictors and often perform well with limited sample sizes, whereas DNN may achieve higher accuracy when large training datasets are available but at the cost of reduced interpretability and higher computational demand. Cubist models, by combining rule-based and linear regression approaches, offer a compromise between prediction accuracy and interpretability. Consequently, no single algorithm can be considered universally superior; rather, model performance depends on data availability, spatial scale and environmental heterogeneity. For example, tree-based ensemble methods tend to outperform linear models in heterogeneous landscapes, while their advantage decreases in more homogeneous conditions.

Model validation is a critical step for assessing the reliability and applicability of SOC predictions. Although split-sample validation and cross-validation are widely used, they may overestimate model performance when training and test data are spatially close. In such cases, models can partially memorize local patterns, leading to high R^2 values while producing spatially unrealistic maps. External validation with independent datasets is therefore essential to better assess generalization ability (Gasmi et al., 2022; Li et al., 2023). Moreover, commonly used metrics such as RMSE and R^2 provide only partial information, as prediction uncertainty often increases in areas with extreme SOC values or poor data coverage (Kumar et al., 2023). Issues of uncertainty and generalizability thus remain major limitations of current SOC mapping studies.

Despite notable progress, several challenges persist. ML models remain highly dependent on the quality, representativeness and spatial distribution of soil observations, which restricts their applicability in data-scarce regions. Furthermore, the predominance of purely data-driven approaches raises concerns regarding model interpretability and transferability across regions and environmental conditions. High predictive accuracy does not necessarily imply process realism, and models trained under specific climatic or land-use contexts may perform poorly when extrapolated beyond their calibration domain.

Future research should therefore move toward integrating multi-source and multi-temporal datasets (e.g., hyperspectral, radar and LiDAR data) to better capture both surface and subsurface SOC controls. The development of process-informed and explainable ML frameworks would improve understanding of the mechanisms underlying SOC variability and enhance confidence in predictions. In addition, standardized validation protocols and explicit uncertainty quantification are required to ensure comparability between studies and to support practical applications. These advances are essential for producing robust and transferable SOC maps that can effectively support long-term monitoring, climate change mitigation strategies and sustainable land management.

6. Conclusions

In response to the global challenge of climate change, scientific advancements offer promising avenues for monitoring soil health and identifying degraded regions through SOC mapping. By quantifying carbon stocks in soils, SOC mapping supports efforts in soil-based carbon sequestration and contributes to climate change mitigation. It also enables the detection of vulnerable areas, guiding land conservation strategies. Moreover, precision SOC mapping empowers farmers to optimize resource inputs, reduce waste, and implement improved soil management practices, thereby enhancing both environmental sustainability and agricultural productivity. The reviewed literature highlights the significant role of ML and RS technologies in SOC prediction. This contribution stems from the rich spectral information captured by multispectral and hyperspectral sensors, and from the capacity of ML algorithms to model complex, nonlinear relationships between environmental variables and SOC spectral signatures. These integrated approaches have proven effective in generating robust, scalable models for accurate SOC estimation across diverse landscapes.

The challenges associated with climate change mitigation, together with the demonstrated effectiveness of recent technological advancements, have made the integration of remote sensing (RS), geographic information systems (GIS), and machine learning (ML) methods a necessity for soil organic carbon (SOC) prediction and mapping. Evidence from both global and national studies highlights the substantial potential of these integrated approaches, while also revealing a number of critical challenges. One major limitation concerns data quality, as the accuracy of SOC predictions depends strongly on the reliability of input datasets; noise, inconsistencies, or errors in environmental variables can significantly degrade ML model performance. In addition, data scarcity remains a persistent issue in regions with limited soil observations, making it difficult to develop robust and transferable models. Although ML techniques enable spatial extrapolation, insufficient or unevenly distributed training data can lead to unreliable predictions. Furthermore, models trained on highly localized datasets may suffer from overfitting, whereby they exhibit strong performance during training but fail to generalize to previously unobserved locations. Collectively, these challenges continue to constrain researchers' ability to fully interpret complex environmental processes, ranging from biodiversity patterns and soil structure dynamics to plant growth and agricultural productivity.

In conclusion, machine learning (ML) algorithms have proven to be effective tools for predicting and mapping SOC content, offering improved accuracy, spatial resolution, and scalability compared with traditional methods. However, ongoing research is needed to address challenges related to model generalization across different regions and to improve the interpretability of model results. With further advancements, these tools can significantly contribute to land use and land cover management as well as carbon sequestration efforts at both global and national levels.

Acknowledgments: I would like to thank the anonymous reviewers and editors for their insightful feedback and constructive suggestions, which greatly contributed to improving this article.

Author Contributions:

E. Khaddi: Conceptualization, Designed the research methodology, Writing—original draft preparation, Writing—reviewing and editing.

H. Zaher: Conceptualization, Designed the research methodology, Writing—original draft preparation, Writing—reviewing and editing.

H. Naim: Conceptualization, Designed the research methodology, Writing—original draft preparation, Writing—reviewing and editing.

AI Contribution Statement: No artificial intelligence (AI) tools were used in the preparation of this manuscript. All work was conducted entirely by the authors.

Funding: This study received no external funding.

Research and publication ethics statement: In the study, the authors declare that there is no violation of research and publication ethics and that the study does not require ethics committee approval.

Conflicts of Interest: The authors declare no conflicts of interest.

Data Availability: The data presented in this study are available on request from the corresponding author.

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